RNS INSTITUTE OF TECHNOLOGY

Autonomous Institution Affiliated to VTU
RR Nagar, Bengaluru-560098

Estd : 2001

II Semester B.E. Degree Examination-June-2026

Computer Aided Engineering Drawing for CSE-Stream

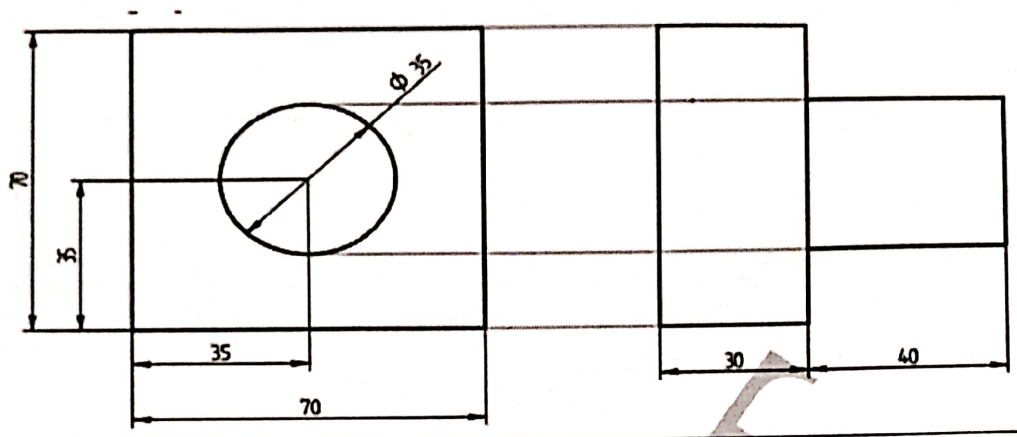
Time: 3 hrs **Max. Marks: 100**

Note:

1. ***Answer any FOUR full questions, choosing at least ONE question from each MODULE.***
2. ***Missing data may be suitably assumed.***
3. ***M: Marks L: Bloom's level, C: Course Outcomes***

		Module - 1	M	L	C
Q. 1	a	A point P is 35mm below HP, 20 mm behind VP and 25 mm from RPP. Draw all the three views of the point.	8	3	1
	b	The top view of a straight line pq is 70 mm and makes an angle of 60^0 with XY line. The end Q is 10 mm in front of VP and 30 mm above HP. The difference between the distances of P and Q above HP is 45mm. draw the projections. Determine its true length and true inclinations with HP and VP.	12	3	1
		OR			
Q. 2	a	A circular lamina of 30mm diameter rests on HP such that one of its diameters is inclined at 45^0 to VP and 30^0 to HP. Draw its top and front views in this position.	20	3	1
		Module – 2			
Q. 3	a	A square pyramid 30 mm sides of base and 60 mm axis length rests on HP on one of its base corner such that the two base edges containing the corner on which it rests makes equal inclination with HP. Draw the projections of the pyramid when the axis is inclined to HP at 40^0 and VP at 30^0.	30	3	2
		OR			
Q. 4	a	A pentagonal prism 25 mm sides of base and 60 mm axis length rests on HP on one of its base corner, such that the two base edges containing the corner on which it rests makes equal inclination with HP. Draw the projections of the prism when the axis is inclined to HP at 40^0 and inclined to VP at 30^0.	30	3	2
		Module – 3			
Q. 5	a	A square prism of base side 30mm and axis length 60mm is resting on HP on its base with all vertical faces being equally inclined to VP. It is cut by an inclined plane 60° to HP and perpendicular to VP and is passing through a point on the axis at a distance 15 mm from the top face. Draw the development of the lower portion of the prism.	25	3	3

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		OR			
Q. 6	a	Following figure shows the front and side views of a solid. Draw the isometric projection of the solid. 	25	3	3
		Module – 4			
Q.7	a	Illustrate the 2D diagram for a ring topology connecting five computers with wired connections. Indicate directionality (use arrows if required).	25	3	4
		OR			
Q.8	a	Draw the 3D orthographic views of an Arduino-board indicating the USB port, power jack, IC, and pin headers.	25	3	4

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Autonomous Institution Affiliated to Visvesvaraya Technological University, Belagavi
Approved by AICTE, New Delhi, Accredited by NAAC with 'A+' Grade
Channasandra, Dr.Vishnuvardhan Road, Bengaluru-560098
Ph:(080) 28611880,28611881URL:www.rnsit.ac.in

Estd - 2001

DEPARTMENT: MECHANICAL ENGINEERING

I INTERNAL TEST

SEM: 2nd SEM -DS-B Sec
SUB: Computer Aided Engineering Drawing
SUB CODE: BCDCS203

Date: 04.06.2026
Time: 3.00– 4.30pm
Max. Marks: 50

Instructions to Candidates:
1Answer all the questions
2. Missing data is suitably assumed

Q. No.	Question	Marks	BC Level	CO's
1	A hemisphere of 40mm diameter is supported co-axially on the vertex of a cone of base diameter 60mm and axis length 50mm. The flat circular face of the hemisphere is facing upside. Draw the isometric projection of the combination of solids.	25	L3	CO3
	OR			
2	The frustum of a square pyramid of sides 40mm, top 20 mm and height 60mm rests on the centre of the top of a square block of side 60 mm and height 20mm. The base edges of the pyramid are parallel to the top edges of the square block. Draw the isometric projection of the combination of the solids	25	L3	CO3
3	Draw a 2D bus topology for six computers using standard symbols. Show and label the backbone cable and devices.	25	L3	CO4
	OR			
4	Draw the 3D orthographic views of an Arduino Uno board indicating the USB port, power jack, IC, and pin headers.	25	L3	CO4

Blooms Cognitive Level:*Remember, Understand, Apply, Analyze, Evaluate, Create*

CO3 - Create isometric projections of simple and combined solids &develop the lateral surfaces of the solids.
CO4 -Create 2-D Drafting of Simple Machine Components/ Devices.

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Vision: Building to become a center of excellence in Mechanical Engineering and to nurture technically competent and socially responsible engineering professionals.

CIE – Test 1

Sem: II-DS-B Sec
Subject: Computer Aided Engineering Drawing (C-Cycle)
Subject Code: BCDCS203

Date:14/05/2026
Time: 2:00PM-3:30PM
Max. Marks: 50

Instruction: Answer any 2 full questions, selecting ONE question from each part.

Q. No.		Question	Marks	RBT	CO
1	a	A point C is on VP, 40mm below HP and 50mm from RPP. Draw its projections.	08	L3	CO
	b	A line AB 80mm long has its end A 20mm above HP and 30mm in front of VP. It is inclined at 30° to HP and 45° to VP. Draw the projections of the line and find apparent lengths and apparent inclinations. Also determine the distance of point B from both HP and VP.	12	L3	CO
		OR			
2	a	A circular lamina of 50mm diameter rests on HP such that one of its diameters is inclined at 30° to VP and 45° to HP. Draw its top and front views in this position.	20	L3	CO
3	a	A cube of 40mm sides rests on HP on an edge which is inclined to VP at 30^0. Draw the projections when the lateral square face containing the edge on which it rests makes an angle of 50^0 to HP.	30	L3	CO
		OR			
4	a	A square pyramid 35mm sides of base and 50mm axis length rests on HP on one of its base corners such that the two base edges containing the corner on which it rests makes equal inclination with HP. Draw the projections of the pyramid when the axis is inclined to HP at 40° and VP at 30°.	30	L3	CO

*** Revised Bloom's Taxonomy: L1-Remember, L2-Understand, L3-Apply, L4-Analyze, L5-Evaluate L6-Create**

Course Outcome:

CO1	Create accurate orthographic projections of points, lines, and planes in the first quadrant, and solve related projection problems.
CO2	Create orthographic projections of prisms, pyramids, cones, cubes, and tetrahedrons resting on the horizontal plane.

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